

An Ultra-Low Power 900 MHz Intermediate Frequency Low Noise Amplifier For Low-Power RF Receivers

Aasish Boora¹, Bharatha Kumar Thangarasu² and Kiat Seng Yeo³
Pillar of Engineering Product Development
Singapore University of Technology and Design
Singapore

aasish_boora@mymail.sutd.edu.sg¹ bharatha_kumar@sutd.edu.sg² kiatseng_yeo@sutd.edu.sg³

Abstract—The recent advancement in biomedical and healthcare sectors shows that the portable ambulatory medical devices with very low power consumption play an important role in continuous monitoring and diagnosis of outpatients by mitigating undesired frequent replacing or recharging of power supply source. To aid this requirement, the fully integrated on-chip circuits should consume very little power without compromising on the overall system performance. In this paper, we present a novel ultra-low power dual-stage intermediate frequency low-noise amplifier (IF LNA) operating at 900 MHz designed in TSMC CMOS 40nm technology. The proposed LNA comprises two identical complementary input stages externally matched to 50 Ω at the load and source along with inter-stage matching. Simulation results of the circuit indicate unconditional stability with a power consumption of 112.9 μ W from a 0.56 V supply, a noise figure of 4.66 dB, and a gain of 10.2 dB. The input-referred IP3 is around -17.2 dBm. This work aims to be incorporated in a fully integrated ultra-low-power (ULP) RF receiver in the 2.4 GHz ISM band.

Keywords—CMOS, integrated circuit, intermediate frequency, low-noise amplifier (LNA), radio frequency receiver, ultra-low power

I. INTRODUCTION

Radio-frequency integrated circuits play a vital role in most of the wireless communication applications. The area of research dealing with the radio frequency front-end (RFFE) has made many significant contributions regardless of major challenges. Some of the challenges include ultra-low power consumption, good sensitivity, and selectivity. The modern-day radio frequency (RF) receivers are consuming large amounts of power to deliver excellent performance. It is very crucial to achieve the desired performance with the least power consumption possible. The reduction of power consumption by a factor of 10-100 would make a huge impact on wireless communications and its wide range of applications. In the existing literature, there are a number of low-noise amplifier (LNA) topologies that are proposed and designed to serve the purpose of providing sufficient gain to the RF signals at the RF front-end with very low power consumption. Some of the ultra-low-power (ULP) LNA topologies are current-reuse gmboosting LNA, inductively degenerated common source LNA, and inverter-based LNA. The main purpose of this work is to design an LNA at 900 MHz frequency which is aimed to be integrated as an IF LNA in the ULP RF receiver operating in 2.4 GHz band with very low power consumption. Architecture

mentioned in [1] is a self-bias inverter with forward body-bias (FBB) operating at 2.45 GHz frequency. It has achieved a gain of 24.4 dB and a noise figure of 2.2 dB with very low power consumption of 12.5 μ W. Architecture mentioned in [2] is a current-reuse common-gate LNA operating at 2.4 GHz frequency. It has achieved a gain of 14.2 dB and a noise figure of 3.3 dB with a power consumption of 30 μ W. Architecture mentioned in [3] is also an inverter-based LNA at 2.4 GHz frequency without FBB. It has achieved a gain of 13.1 dB and a noise figure of 5.3 dB consuming 60 μ W power. Similar topology with inductive shunt-feedback is used in [4] for ultra-wideband applications. When it comes to LNA design at lower frequencies, the above mentioned topologies should be carefully implemented to achieve a good gain with very low power consumption. LNAs mentioned in [5], [6] and [7] are operated in sub-GHz frequency range. In [5], multi-gate common source topology is used to improve nonlinearity of the LNA and the design has achieved a gain of 14.9 dB and a noise figure of 1.8 dB with a large power consumption of 5.5 mW. In [6], an inductively degenerated LNA topology is used to achieve a gain of 17.5 dB and a noise figure of 2 dB with a power consumption of 21.6 mW.

This paper tries to design an LNA at 900 MHz frequency adapting some of the ultra-low-power LNA topologies mentioned above using the Cadence tool. The proposed ULP LNA architecture helps to understand the challenges involved in designing a specific LNA topology at lower frequencies. This work has achieved a gain of 10.2 dB and a noise figure of 4.66 dB at a very low power consumption of 112.9 μ W. The existing ULP LNA architectures and their advantages are discussed in Section II. The design methodology and circuit description of

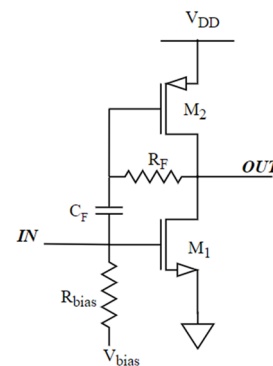


Fig. 1. Inverter based LNA adapted from [1].

the proposed LNA are mentioned in Section III. Section IV presents the results obtained from the pre- and post-layout simulations. Finally, Section V briefs the importance of the designed LNA and concludes the work.

II. EXISTING ULP LNA TOPOLOGIES

From the existing literature on LNAs, two topologies are often used to design LNA for ULP applications. One is the self-biased inverter-based LNA and the other one is cross-coupled common gate (CCCG) LNA. An intuitive approach to reduce power consumption is lowering the supply voltage. Operating the transistors in saturation region with a very low voltage headroom can be achieved by lowering the threshold voltage of cascoded transistors. To achieve threshold voltage lowering, MOSFET body-to-source junctions can be forward-biased which is known as a forward body-bias (FBB) technique. Usually, this method is incorporated in inverter-based LNA topologies to operate the designed LNAs at supply voltages as low as 0.35 V [1]. Some of the referred ULP LNA topologies are shown in the figures Fig. 1, Fig. 2, and Fig. 3.

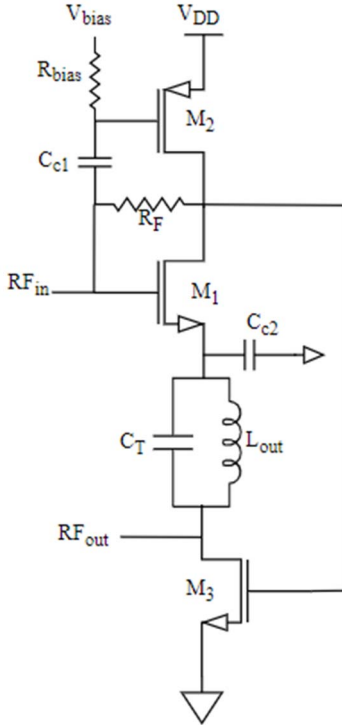


Fig. 2. Current-reuse gm-boosting LNA adapted from [8] and [9].

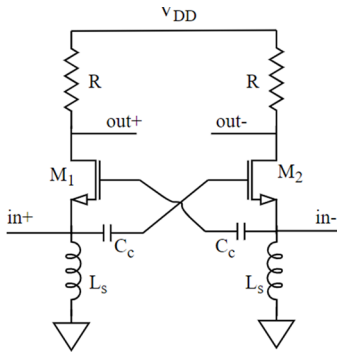


Fig. 3. Cross-coupled common gate LNA adapted from [11].

III. METHODOLOGY AND CIRCUIT DESIGN OF PROPOSED LNA

A. Design Flow

The performance expectations and desired specifications for various parameters of an LNA are decided prior to the design. The first step is to choose a suitable LNA topology that has the potential to achieve the desired performance. The chosen LNA topology is implemented at the circuit level for performance analysis. Various simulations will be performed to evaluate all the parameters required to analyze the circuit performance called the pre-layout simulations. Once the simulation results are satisfactory, the second step is to draw a layout for the designed circuit schematic carefully with minimum losses in the signal path. The third step is to check the layout compliance with design rules. Layout design should undergo design-rule check (DRC) and the layout will be compared with the schematic for exact mapping, called the layout versus schematic (LVS) check. The final step will be to perform parasitic extraction and execute post-layout simulations to analyze the circuit performance. The detailed design methodology is illustrated using flow-chart in Fig. 4.

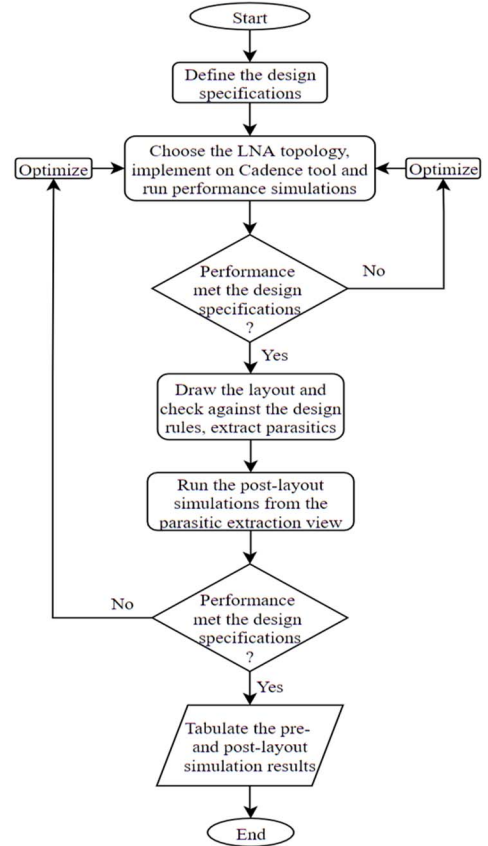


Fig. 4. Design methodology flow-chart.

B. Circuit Design

The proposed LNA is designed for ultra-low power applications. To achieve all the desired specifications with very low power consumption, the LNA topology must be chosen carefully. Based on the literature study on LNAs, the inverter-based shunt feedback LNA is found to be more suitable for ultra-low power applications. Such an LNA topology is implemented

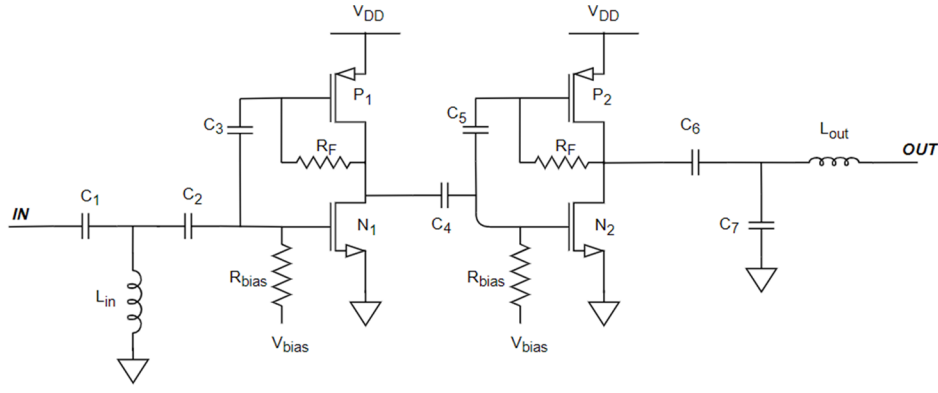


Fig. 5. Proposed LNA circuit schematic.

in two stages efficiently in this work. The input matching network contains an inductor and two capacitors forming a T-network. The output matching network also includes an inductor and two capacitors. The inverter-based LNAs are cascaded one after the other with proper inter-stage matching. The circuit component values are mentioned in Table I. The LNA circuit schematic along with input and output matching networks are shown in Fig. 5.

TABLE I. CIRCUIT COMPONENT DESCRIPTION TABLE

Component/variable	Value
C_1	710 fF, $Q = 797.5$
L_{in}	37.074 nH, $Q = 8.67$
C_2	1.291 pF, $Q = 897.6$
C_3	1.035 pF, $Q = 961$
R_{bias} and R_F	60 K Ω
N_1 and N_2	28 μ m/60 nm
P_1 and P_2	80 μ m/50 nm
C_4	1.211 pF, $Q = 693.5$
C_5	517.8 fF, $Q = 961$
C_6	3.1 pF, $Q = 961$
C_7	704 fF, $Q = 809$
L_{out}	32.96 nH, $Q = 11.21$

C. Test Bench

The test bench used to perform the pre-layout simulations on the circuit schematic is shown in Fig. 6. The symbol for the LNA

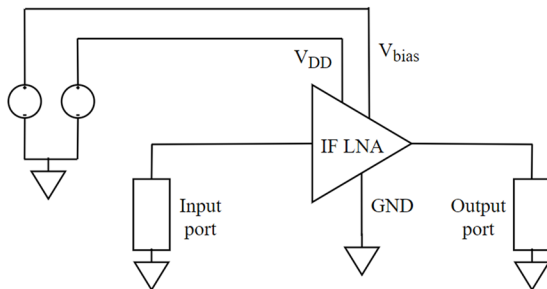


Fig. 6. LNA simulation test bench.

schematic is created to use it in the test bench. Only two voltage sources are required for the LNA. One is the power supply voltage source and the other is bias voltage given to the NMOS transistor. While doing an overall system fabrication on-chip, the bias voltage can be generated from the supply voltage with the help of a voltage generator circuit. The total current consumed by the LNA is around 204 μ A at a voltage supply of 0.56 V.

IV. SIMULATION RESULTS AND DISCUSSION

A. Pre-layout Simulation Results

The designed LNA achieved a peak gain of 12.09 dB at the operating frequency of 900 MHz with a 1 dB bandwidth of 115 MHz. The input and output matching networks are perfectly designed to meet the desired specifications. The overall power consumption reported by DC simulation is around 114.2 μ W a little exceeding the limit (100 μ W). The LNA is unconditionally-stable over the entire frequency range from DC to 10 GHz. The parameters considered for stability analysis in the s-parameter simulation are K_f and B_{lf} .

B. Post-layout Simulation Results

Before proceeding to execute the performance-based simulations on the layout, we need to ensure that the drawn layout is foundry compatible (design-rule check), and is an exact match with the earlier designed schematic (Layout vs schematic check). Electrical rule check (ERC) is also an important aspect that the layout needs to pass in order to ensure all the well and substrate connections with proper contacts and spacing. Parasitic elements such as stray capacitance, coupling capacitance, and distributive resistance can be extracted using various features supported by the Cadence tool. In this project, we have used Calibre-PEX for parasitic extraction. The designed LNA achieved a gain of 10.2 dB in post-layout simulations at 900 MHz frequency. The input and output matching networks are properly designed to minimize signal reflections at both ends. The overall power consumption is around 112.9 μ W evaluated from the post-layout DC simulation. The LNA is unconditionally stable over a wide range of frequencies i.e., from DC to 10 GHz. The pre- and post-layout simulation results

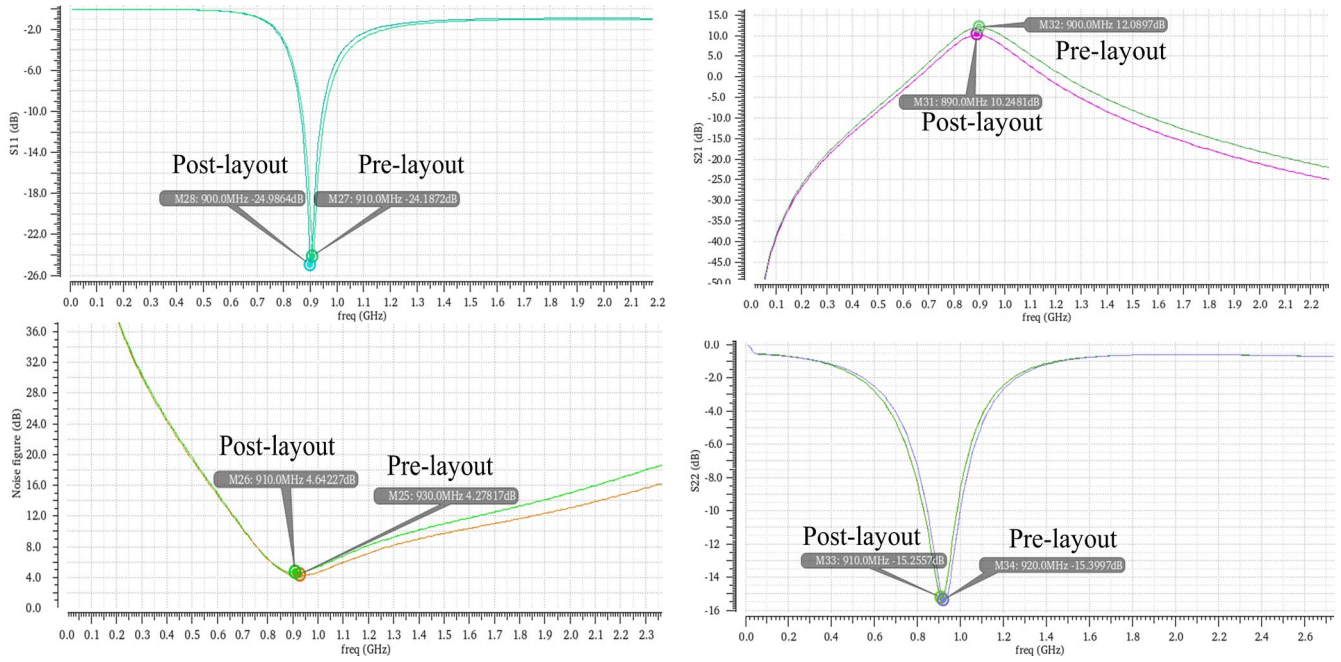


Fig. 7. Simulated pre- and post-layout performance comparison plot.

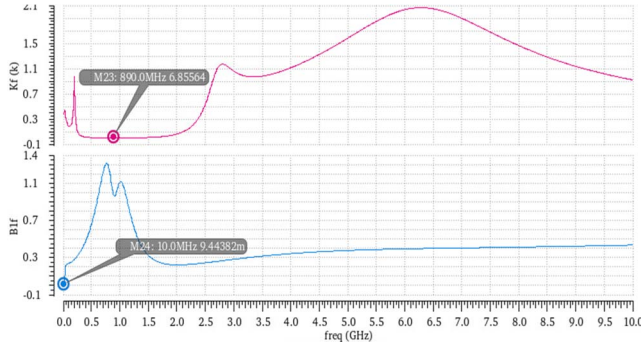


Fig. 8. Simulated plot of stability factors K_f and B_{1f} post-layout.

with comparison plots are shown in Fig. 7 and the stability plots are shown in Fig. 8.

The pre- and post-layout performance of the designed IF LNA circuit is tabulated as shown below in Table II. There is a large variation between pre- and post-layout performance with respect to some parameters such as S_{11} and S_{12} . This discrepancy is due to the parasitics at the input port affecting the minima point of S_{11} . Since the S_{11} plot is narrow, even a small shift in minima point could result in a larger deviation in parameter value at the center frequency. At a center frequency of 900 MHz within a bandwidth of 2.5 MHz, the LNA occupied an area of 0.664 mm². The layout of the designed IF LNA is shown in Fig. 9.

Some of the novel ULP architectures are considered to perform a comparison against various parameters in this work. To do a fair comparison, LNA designs operating at the same frequency as this work, which is 900 MHz, are taken into consideration. The proposed LNA achieved sufficient gain at very low power consumption and is comparable with the performance of other published LNAs. The parameters of this

work that are considered for comparison with state-of-the-art LNA topologies are shown in Table III.

TABLE II. PRE- AND POST-LAYOUT PERFORMANCE COMPARISON TABLE

Parameter	Pre-layout results within 2.5 MHz bandwidth	Post-layout results (R+C+CC) within 2.5 MHz bandwidth
Frequency (MHz)	900	
Area (mm ²)	0.664 (0.975 mm X 0.681 mm)	
Power dissipation (μ W)	114.3	112.9
Power gain (S_{21} in dB)	≥ 12.08	≥ 10.21
Noise figure (dB)	≤ 4.37	≤ 4.66
S_{11} (dB)	≤ -19.35	≤ -24.18
S_{12} (dB)	≤ -44.42	≤ -33.20
S_{22} (dB)	≤ -14.30	≤ -14.92
Input referred IP_3 (dBm) ^a	-19.46	-17.21
Input referred P_{1dB} (dBm) ^b	-38.54	-36.88
Stability	$K_f > 1$ & $B_{1f} > 0$ from DC to 10 GHz	$K_f > 1$ & $B_{1f} > 0$ from DC to 10 GHz

^a Third-order intermodulation product between signal tones 5 MHz apart centered at 900 MHz.

^b Mentioned value is for signals at 900 MHz frequency.

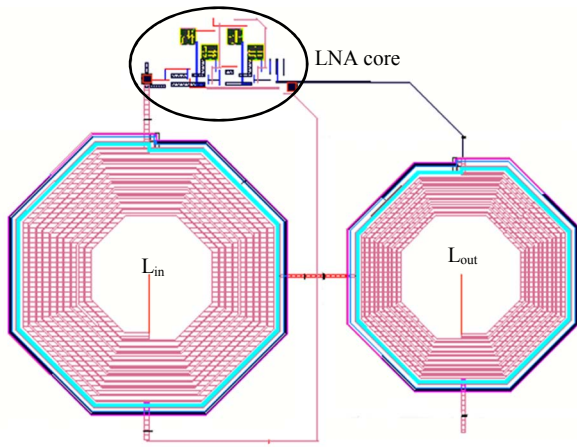


Fig. 9. Layout of IF LNA (total area = 0.664 mm²).

TABLE III. PERFORMANCE COMPARISON OF PUBLISHED LNAs

	This work	[5]	[10]	[11]
Technology	40 nm	180 nm	90 nm	180 nm
Frequency (MHz)	900	900	40 to 800	80 to 800
Bandwidth (MHz) ^a	200	Not given	760	720
Topology	Dual-stage Inverter	Multi-gate common-source	Current-reuse common-gate	CCCG
Balance	Single-ended	Single-ended	Fully-differential	Fully-differential
S ₂₁ (dB)	10.2	14.9	14.27	17.2
NF (dB)	4.66	1.8	6.5	5.4
IIP ₃ (dBm)	-17.21	9.4	-5.74	-11.85
Power (μW)	112.9	5550	490	200
VDD (V)	0.56	1.5	1.2	1

^a 3 dB bandwidth.

V. CONCLUSION

This paper presents an IF LNA designed to operate at 900 MHz frequency with very low power consumption. The designed IF LNA employs inverter-based shunt-feedback configuration. Two such stages are cascaded together to achieve good performance. The designed LNA is operated at a supply

voltage of 0.56 V and consumed a power of 112.9 μW. The designed LNA has a simulated power gain of 10.2 dB and a noise figure of 4.66 dB. The inverter-based LNA topology is chosen to reduce power consumption. This proposed LNA can be incorporated in a fully integrated ultra-low-power RF receiver in the 2.4 GHz ISM band.

ACKNOWLEDGMENT

This project is supported by Singapore's National Research Foundation, Grant Award Number: NRF-CRP20-2017-0006.

REFERENCES

- [1] X. Aragones et.al., "Design of ULV ULP LNAs exploiting FBB in FDSOI 28nm technology," in *XXXIV Conf. Design of Circuits and Integr. Syst. (DCIS)*, Bilbao, Spain, Nov. 20-22, 2019, pp. 1-6.
- [2] E. Kargaran, D. Manstretta, and R. Castello, "Design and analysis of 2.4 GHz 30 μW CMOS LNAs for wearable WSN applications," *IEEE Trans. Circuits and Syst. I, Reg. Papers*, vol. 65, no. 3, pp. 891-903, Mar. 2018.
- [3] T. Taris, J. Begueret, and Y. Deval, "A 60μW LNA for 2.4 GHz wireless sensors network applications," in *IEEE Radio Freq. Integr. Circuits Symp. (RFIC)*, Baltimore, MD, June 5-7, 2011, pp. 1-4.
- [4] M. Parvizi et.al., "A Sub-mW, ultra-low-voltage, wideband low-noise amplifier design technique," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 23, no. 6, pp. 1111-1122, June 2015.
- [5] T. H. Jin and T. W. Kim, "A 5.5-mW +9.4-dBm IIP₃ 1.8-dB NF CMOS LNA employing multiple gated transistors with capacitance desensitization," *IEEE Trans. Microw. Theory Tech*, vol. 58, no. 10, pp. 2529-2537, Oct. 2010.
- [6] F. Gatta et.al., "A 2-dB noise figure 900-MHz differential CMOS LNA," *IEEE J. Solid-State Circuits*, vol. 36, no. 10, pp. 1444-1452, Oct. 2001.
- [7] L. Liu, T. Sakurai, and M. Takamiya, "315MHz energy-efficient injection-locked OOK transmitter and 8.4μW power-gated receiver front-end for wireless ad hoc network in 40nm CMOS," in *Symp. on VLSI Circuits Dig. Tech. Papers*, Honolulu, HI, June 15-17, 2011, pp. 164-165.
- [8] A. Dissanayake et.al., "A 64 μW, 23 dB gain, 8 dB NF, 2.4 GHz RF front-end for ultra-low power internet-of-things transceivers," in *IEEE Radio Freq. Integr. Circuits Symp. (RFIC)*, Honolulu, HI, June 4-6, 2017, pp. 184-187.
- [9] K. Kumar et.al., "A 75-μW 2.4 GHz wake-up receiver in 65-nm CMOS for neonatal healthcare application," in *Int. Conf. VLSI Design (VLSID)*, Delhi, India, Jan. 5-9, 2019, pp. 287-292.
- [10] W. Chen, S. Yang, and K. Cheng, "A 1.2 V 490 μW sub-GHz UWB CMOS LNA with current reuse negative feedback," in *IEEE Int. Symp. Circuits and Syst. (ISCAS)*, Florence, May 27-30, 2018, pp. 1-4.
- [11] Marzuki et.al., "A differential cross-coupling common gate low noise amplifier (LNA) for medradio band application," *Int. J. Elect. Electron. Eng. & Telecommun.*, vol. 9, no. 1, Jan. 2020.